Polaris Vega and Lyra Firmware Revision History

10001162 R12
May 2023

Revision Status

Revision Number	Date	Description
01	September 2017	First release
02	April 2018	Updated for firmware package 004.
03	June 2018	Updated the section "Improvements and Modifications" on page 53. Added the command “GET Features.Firmware.Package Number” to Table 1-1 on page 1.
04	June 2019	Updated for firmware package 006.
05	October 2019	Updated for patches to firmware versions 003 and 006.
06	February 2020	Updated for firmware package 007.
07	March 2020	Updated for package 006.022.
08	October 2020	Updated for package 006.023.
09	December 2021	Updated for package 008.012.
10	March 2022	Updated for package 008.013.
11	December 2022	Updated for package 009.011. Included information about changes made to the `Info.Status.Gravity Vector` parameter in package 008.012 and the bug fix made for the ST/VT Exposure Control issue in package 007.011.
12	May 2023	Updated for package 010.017. Included information about Polaris Lyra support, GPIO functionality for Lyra, and secure host-to-device communication via TLS/DTLS. Updated information for package 008.012 to indicate that trailing space are no longer accepted in API commands.

Part Number: 10001162

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Table of Contents

1 Introduction 1

1.1 API Commands for Firmware Revisions 1

1.2 Firmware Release Versioning Parameters and Rules 1

Combined Firmware Revision 1

Package Number 1

Application Version 2

API Revision 2

1.3 Known Issue 2

Polaris Vega or Lyra May Become Unreachable After Network Interruption 2

2 Vega and Lyra Version 010 Firmware 3

2.1 Package 010.017 3

Package Contents 3

API Version 4

New Features 5

Improvements and Modifications 5

Bug Fixes 6

Known Issues 7

3 Vega Version 009 Firmware 8

3.1 Package 009.011 8

Package Contents 8

API Version 9

New Features 9

Improvements and Modifications 9

Bug Fixes 9

Known Issues 10

4 Vega Version 008 Firmware 11

4.1 Package 008.013 11

Package Contents 11

API Version 12

New Features 12

Improvements and Modifications 12

Bug Fixes 12
Known Issues 13
4.2 Package 008.012 14
Package Contents 14
API Version. 15
New Features 17
Improvements and Modifications 17
Bug Fixes 18
Known Issues 20

5 Vega Version 007 Firmware 21
5.1 Package 007.011 21
Package Contents 21
API Version. 22
New Features 23
Improvements and Modifications 23
Bug Fixes 23
Known Issues 24

6 Vega Version 006 Firmware 25
6.1 Package 006.023 25
Statement of Compliance 25
Package Contents 25
API Version. 26
New Features 26
Improvements and Modifications 27
Bug Fixes 27
Known Issues 27
6.2 Package 006.022 28
Statement of Compliance 28
Package Contents 28
API Version. 29
New Features 29
Improvements and Modifications 30
Bug Fixes 30
Known Issues 30

6.3 Package 006.021 31
Statement of Compliance 31
Package Contents 31
API Version. 32
New Features 32
Improvements and Modifications 32
Bug Fixes 33
Known Issues 33
6.4 Package 006.019 34
Package Contents 34
API Version. 35
New Features 36
Improvements and Modifications 37
Bug Fixes 38
Known Issues 39

7 Vega Version 004 Firmware 40
7.1 Package 004.008 40
Package Contents 40
API Version. 40
New Features 41
Improvements and Modifications 42
Bug Fixes 42
Known Issues 43

8 Vega Version 003 Firmware 44
8.1 Package 003.017 44
Statement of Compliance 44
Package Contents 44
API Version. 45
New Features 45
Improvements and Modifications 45
Bug Fixes 46
Known Issues 46
8.2 Package 003.016 47
Statement of Compliance 47

Package Contents 47
API Version. 47
New Features 48
Improvements and Modifications 48
Bug Fixes 48
Known Issues 49
8.3 Package 003.015 50
Package Contents 50
API Version. 50
New Features 53
Improvements and Modifications 53
Bug Fixes 54
Known Issues 55

9 Vega Version 002 Firmware 57
9.1 Package 002.005 57
Package Contents 57
API Version. 57
New Features 59
Improvements and Modifications 59
Bug Fixes 59
Known Issues 59

10 Polaris Vega and Lyra Boot and Safe Loader. 61

11 Abbreviations and Acronyms 62

1 Introduction

This guide provides a reference of the firmware revision history for the Polaris Vega and Lyra Systems. It details the history of enhancements and fixes made to the Vega and Lyra firmware components.

1.1 API Commands for Firmware Revisions

To query API and firmware revisions, use the following commands:

Table 1-1 Command List

Command	Description
`GET Features.Firmware.API Revision`	Preferred method to return the API revision number. The API revision number defines the set of commands and parameters available to communicate with the system.
`APIREV`	Deprecated. Returns the API revision number.
`GET Features.Firmware.Combined Firmware Revision`	Returns the Combined Firmware Revision.
`GET Features.Firmware.Package Number`	Returns the package number of the loaded and active firmware.
`VER 5`	Returns the Combined Firmware Revision (not recommended, only for backward compatibility).
`GET Features.Firmware.*`	Returns additional (component level) revision information such as Boot Loader, Safe Loader and application firmware.

For further information on the API, see the *Polaris Vega and Lyra Application Programming Interface Guide.*

1.2 Firmware Release Versioning Parameters and Rules

Combined Firmware Revision

The Combined Firmware Revision value, available via the `Features.Firmware.Combined Firmware Revision` parameter, indicates the general revision of firmware that is executing on the Polaris Vega and Lyra optical tracking systems and represents a specific set of features, improvements, and bug fixes. NDI increments this value with each new significant firmware release, normally in association with a new set of features, improvements, and bug fixes. The set of new features introduced in a new Combined Firmware Revision may include support for new hardware and associated enhanced functionality that may offer further benefit to customers.

Package Number

The firmware Package Number, available via the `Features.Firmware.Package Number` parameter, takes the form 007.011 and indicates both the Combined Firmware Revision (007) and the package build number (011). To provide uniqueness, NDI increments the build number for each release of firmware.

NDI aims to limit the number of firmware builds for each Combined Firmware Revision; however, NDI might need to occasionally publish a patch release to correct critical bugs found in a particular Combined Firmware Revision. If this happens, all packages with the same Combined Firmware Revision number will contain the same functionality; the only difference will be the additional bug fixes in the packages with higher build numbers.

Application Version

The version of the firmware application executing on the Polaris Vega and Lyra optical tracking systems, available via the `Features.Firmware.Version` parameter, takes the form 008.007.011 and indicates the major version (008), minor version (007), and build number of the application (011).The major version is always 008 for Vega and Lyra Systems. The major version, minor version, and build number are also available via the `Features.Firmware.Major Version`, `Features.Firmware.Minor Version`, and `Features.Firmware.Build Number` parameters respectively.

API Revision

The API revision of the Vega and Lyra firmware, available via the `Features.Firmware.API Revision` parameter, takes the form X.003.005, where X is ‘G’ for the Position Sensor and System Control Unit and ‘V’ for the VCU. This parameter revisions the “language” of the Vega and Lyra device API, which includes the set of both commands and parameters. Note that the Application Version described above revisions the “behaviour” of the Vega and Lyra devices.

The major version (003) is incremented for changes that are not backwards compatible, and the minor version (005) is incremented for changes that are backwards compatible.

Note **Typically, the concatenation of the firmware application minor version and build number is equivalent to the firmware Package Number. However, this is not always the case. For example, if a “patch release” for Combined Firmware Revision 007 is necessary due to a required FPGA functionality update, the Combined Firmware Revision number and the application version would not need to be incremented, but the build number in the firmware Package Number would need to be incremented by one. Using the example of 008.007.011, the Combined Firmware Revision would remain 007 and the application version would remain 008.007.011, but the build number in the firmware Package Number would be incremented from 007.011 to 007.012.**

1.3 Known Issue

The following known issue is present in all versions of the Vega and Lyra firmware.

Polaris Vega or Lyra May Become Unreachable After Network Interruption

If the network that the Vega or Lyra Position Sensor is connected to is disconnected or goes down and is reconnected without power cycling the Position Sensor, the network connection to the Position Sensor may not be initialized properly. In this case, the Position Sensor can become unreachable. Power-cycling the Position Sensor will restart the network services and then the connection can be re-established.

Note **This problem can occur when the PoE midspan is disconnected and reconnected without power cycling the Position Sensor.**

2 Vega and Lyra Version 010 Firmware

This chapter provides a reference of the history of firmware revision 010 for the Polaris Vega and Lyra Systems. It details the history of enhancements and fixes made to the Vega and Lyra firmware.

Note **Previous versions of firmware detailed in this guide were for the Polaris Vega model only.**

Firmware revision 010 introduces:

- Support for the Polaris Lyra Position Sensor
- Support for the new GPIO functionality introduced with Lyra
- Secure host-to-device communication via TLS/DTLS transport security
- Bug fixes and improvements

2.1 Package 010.017

This is the first release of version 010 firmware.

Package Contents

The package structure was updated in 010.017 to reduce the number of individual firmware components.

Table 2-1 Package Contents - Package 010.017 (March 2023)

Component	Version
Position Sensor File System	10.17
Video Camera File System	10.17
SCU File System	10.17
STB Control Firmware	000.005.001

Table 2-2 Package Contents - Boot Loaders and Safe Loaders

Component	Version
Position Sensor Boot Loader	2023.02.01
Position Sensor Safe Loader	10.17
Video Camera Boot Loader	2018.01.17
Video Camera Safe Loader	10.17
SCU Boot Loader	2023.02.01
SCU Safe Loader	10.17

API Version

Table 2-3 Version Details - Package 010.017

API Revision - Position Sensor	G.003.007
API Revision - Video Camera Unit	V.001.001
API Revision - System Control Unit	G.003.007

New Commands

Firmware version 010 introduces the following command:

- `GENKEY`

For more information, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

Modified Commands

The API command `BX2` now includes the following information via GBF in each response where a 3D Data Component is included:

- Marker Error for markers assigned to tools
- Marker Line Separation data

For more information on the GBF components, see the *Polaris Vega and Lyra Application Programming Interface Guide*. For more information on how to use the Marker Error information when designing and characterizing tools, see the *Polaris Tool Design Guide*.

`GET`, `GETINFO`, and `SET` now consistently return Error Code 7 (incorrect number of parameters) when an incorrect number of parameters is specified for the command.

`SET` now returns Error Code 39 (permission denied) for non-writeable array parameters.

New Parameters

`Device.Connection.<idx>`: Describes how to reach the device using the Address and Port values. Not implemented for Polaris Vega or Polaris Lyra.

`Features.Hardware.Characterization Date`: Provides the characterization data of the Position Sensor. Previously, the date was available in the VER 4 response only.

`Param.Connect.Open Ports`: Selects the socket security mode.

`Param.GPIO.*`: Support the new GPIO functionality introduced with Lyra.

`Param.Network.TLS.*`: Support secure host-to-device communication via TLS/DTLS.

Modified Parameters

`Info.Status.Gravity Vector`: In firmware revision 006, the value of this parameter was reduced to 2 decimal places to avoid giving the impression that the numbers are more precise than

they really are. However, this is insufficient to represent every possible 8-bit increment. The displayed precision is now increased to 4 decimals.

`Param.Clock.Date Time`: Sets the real-time clock (RTC), and now also sets the current system time.

`Info.Status.Alerts`: New alerts introduced. For details, see "New Info.Status.Alerts Alerts" on page 5.

`Param.Tracking.Track Frequency`: For Lyra, valid values are 0 (1/3 the frame frequency) or 1 (1/2 the frame frequency). The maximum track frequency for Lyra is 1/2 the frame frequency.

Reads of write-only parameters now consistently return either “” or 0, depending on the parameter type.

New Features

Lyra GPIO Functionality

This firmware revision introduces support for the new Lyra GPIO functionality. For more information, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

Secure Communication

This package introduces an optional secure data transmission feature for Vega and Lyra. If enabled, data transmitted between the Position Sensor and the host computer (both directions) is encrypted. This protects against tampering of the transmitted data. It does not limit which devices (computers) can receive the data and does not protect against tampering of the firmware.

This feature supports TLS (transport layer security) for TCP connections and DTLS (datagram transport layer security) for UDP connections. For more information, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

Improvements and Modifications

New Info.Status.Alerts Alerts

File System Full: Indicates when the file system is full. An example cause is when too many entries have been written to the device event log. (0x02000000)

RTC Fault: Set when the RTC oscillator stopped fault bit is stuck or when communication with the RTC is unsuccessful. (0x00000400)

GPIO SYNC IN out of range: Set when the external frequency provided via the GPIO SYNC IN signal deviates more than 10% from the nominal frequency. (0x00008000)

Password Protection

Previously, it was possible to overwrite a set `Config.System Password` without first authenticating through `Config.Password`. This is now addressed. Further, the password string set to `Config.System Password` must be at least 15 characters long and you can no longer set a system password via a feature key.

Note **If you forget the set `Config.System Password`, you must send the device back to NDI for recovery.**

Unsupported Feature Keys

Enabling an unsupported feature key now raises a recoverable parameter fault (0x01000000).

Bug Fixes

Unsuccessful Tool Lock-On

In previous releases, a Vega XT at 325 Hz +/- 20 Hz frame frequency, and with a frame sequence of two in length, may not have successfully locked on to a tool positioned statically in the volume.

Permanent Block of Master Connection

In previous releases, when host communication to the Vega was broken in the middle of receiving a command response, another new connection to the Vega would not be able to become master and would remain in monitor mode. A device reboot was required to recover from this state. Firmware revision 010 fixes this issue.

VCAP Response Trigger Threshold

This release fixes a problem on Vega XT where the Trigger Threshold value in the `VCAP` response reported an incorrect value.

System Control Unit Strober Firmware Improvements

Improvements were made to the System Control Unit strober firmware to ensure that manufacturing parameters retain correct values. This release increments the strober firmware version to 000.005.001.

Timeout when Data Averaging Depth Greater than 8

Previously, a `BX2` command timeout would occur when the data averaging depth was set greater than 8 and the frame sequence was two or more.

Video Camera Unit Improvements

`RTSP` now properly handles port number roll over upon reconnect.

Previously, setting `Param.Exposure And Gain.Auto` to **0** could result in a Video Camera Unit watchdog reset.

Previously, the Auto White Balance algorithm didn’t properly adjust for a fully grey input image.

Repeated RESET 0 Commands

Addressed a problem where a series of back-to-back `RESET 0` commands could cause a device watchdog.

Known Issues

Lyra GPIO Illuminator Enable and Active Wired Tools

For Lyra Position Sensors, if the function of the GPIO2 pin is set to `ILLUMINATOR ENABLE`, tracking of active wired tools is not reliable.

3 Vega Version 009 Firmware

This chapter provides a reference of the history of firmware revision 009 for the Polaris Vega System. It details the history of enhancements and fixes made to the Vega firmware.

Firmware revision 009 introduces:

- new functionality for multi-face tools to track using all visible markers, regardless of the associated face
- bug fixes and improvements

Note **This firmware version has not been verified on the original Vega Position Sensor, sometimes referred to as Vega Classic or Vega 1.0.**

3.1 Package 009.011

This is the first release of version 009 firmware.

Package Contents

Table 3-1 Package Contents - Package 009.011 (November 2022)

Component	Version
Position Sensor Control Application	008.009.011
Position Sensor Driver Firmware	2.35.0
Position Sensor File System	9.11
Position Sensor Firmware Compatibility File	009
Position Sensor Main Board Logic for 7010	211108-124603
Video Camera File System	9.11
SCU Control Application	008.009.011
SCU Driver Firmware	2.35.0
SCU File System	9.11
SCU Firmware Compatibility File	009
SCU Main Board Logic	211129-103732
STB Control Firmware	000.005.000

Table 3-2 Package Contents - Boot Loaders and Safe Loaders

Component	Version
Position Sensor Boot Loader	2019.03.05
Position Sensor Safe Loader	9.11
Video Camera Boot Loader	2018.01.17

Table 3-2 Package Contents - Boot Loaders and Safe Loaders (Continued)

Component	Version
Video Camera Safe Loader	9.11
SCU Boot Loader	2019.03.05
SCU Safe Loader	9.11

API Version

Table 3-3 Version Details - Package 009.011

API Revision - Position Sensor	G.003.006
API Revision - Video Camera Unit	V.001.000
API Revision - System Control Unit	G.003.006

Commands and Parameters

No commands were introduced, modified, deprecated, or removed in this release. No parameters were introduced, modified, deprecated, or removed in this release.

New Features

This package introduces a new tracking mode to track all visible markers on a multi-face tool rather than using only markers assigned to the optimal face. NDI recommends using this new tracking mode as it can improve the accuracy of tool tracking by increasing the number of markers used to compute the tool transformation. This tracking mode is also more robust if markers on the optimal face are occluded.

For more information about this tracking mode, see the *Polaris Tool Design Guide*, Revision 02 or later.

Improvements and Modifications

N/A

Bug Fixes

Position Sensor Reset With Active Wired Tools and UDP Streaming

Package 008.013 had a known issue where tracking active wired tools when streaming tracking data via the UDP protocol sometimes caused a Position Sensor reset when `TSTART` was issued or when an active wired tool was plugged in or unplugged. Package 009.011 fixes this problem.

Position Sensor Reset When All Frame Types are Enabled

Package 008.013 had a known issue where issuing `TSTART` or `VCAP` while tracking passive tools with active wireless tools and with active wired tools, and with illuminated and background frames enabled, caused a Position Sensor reset. Package 009.011 fixes this problem.

Known Issues

N/A

4 Vega Version 008 Firmware

This chapter provides a reference of the history of firmware revision 008 for the Polaris Vega System. It details the history of enhancements and fixes made to the Vega firmware.

Firmware revision 008 introduces:

- enhanced streaming capabilities
- data averaging functionality
- an increased frame rate of up to 400 Hz for Vega XT
- reduced data latency for Vega XT
- active wireless tools for Vega XT
- active wired tools for all Vega models
- bug fixes and improvements

4.1 Package 008.013

This is the second release of version 008 firmware. This package introduces full support for active wired tool tracking.

Package Contents

Table 4-1 Package Contents - Package 008.013 (February 2022)

Component	Version
Position Sensor File System	8.13
Position Sensor Firmware Compatibility File	008
Position Sensor Control Application	008.008.013
Position Sensor Driver Firmware	2.35.0
Position Sensor Main Board Logic for 7010	211108-124603
Video Camera File System	8.13
SCU File System	8.13
SCU Firmware Compatibility File	008
SCU Control Application	008.008.013
SCU Driver Firmware	2.35.0
SCU Main Board Logic	211129-103732
STB Control Firmware	000.005.000

Table 4-2 Package Contents - Boot Loaders and Safe Loaders

Component	Version
Position Sensor Boot Loader	2019.03.05

Table 4-2 Package Contents - Boot Loaders and Safe Loaders (Continued)

Component	Version
Position Sensor Safe Loader	8.13
Video Camera Boot Loader	2018.01.17
Video Camera Safe Loader	8.13
SCU Boot Loader	2019.03.05
SCU Safe Loader	8.13

API Version

Table 4-3 Version Details - Package 008.013

API Revision - Position Sensor	G.003.006
API Revision - Video Camera Unit	V.001.000
API Revision - System Control Unit	G.003.006

Commands and Parameters

No commands were introduced, modified, deprecated, or removed in this release. No parameters were introduced, modified, deprecated, or removed in this release.

New Features

This package introduces full support for active wired tool tracking for all Vega models.

Improvements and Modifications

N/A

Bug Fixes

Position Sensor Reboot with Type 1 PSE

Package 008.012 introduced a problem where a Vega Position Sensor connected to a low-power Type 1 (13 W) PSE might reboot one or more times after boot-up. After multiple reboots, it booted into Safe Mode. Package 008.013 fixes this problem.

Position Sensor Acknowledgment of PSE Power Allocation

LLDP testing identified that the Position Sensor did not reliably acknowledge new power allocations by the PSE, which violates the LLDP protocol defined in clause 33.6.2 of the 802.3at standard. Package 008.013 fixes this problem.

Known Issues

Position Sensor Reset With Active Wired Tools and UDP Streaming

Tracking active wired tools when streaming tracking data via the UDP protocol sometimes causes a Position Sensor reset when `TSTART` is issued or when an active wired tool is plugged in or unplugged.

Position Sensor Reset When All Frame Types are Enabled

Issuing `TSTART` or `VCAP` while tracking passive tools with active wireless tools and with active wired tools, and with illuminated and background frames enabled, causes a Position Sensor reset.

4.2 Package 008.012

This is the first release of version 008 firmware.

Package Contents

Table 4-4 Package Contents - Package 008.012 (December 2021)

Component	Version
Position Sensor File System	8.12
Position Sensor Firmware Compatibility File	008
Position Sensor Control Application	008.008.012
Position Sensor Driver Firmware	2.34.0
Position Sensor Main Board Logic for 7010	211108-124603
Video Camera File System	8.12
SCU File System	8.12
SCU Firmware Compatibility File	008
SCU Control Application	008.008.012
SCU Driver Firmware	2.34.0
SCU Main Board Logic	200305-155548
STB Control Firmware	000.004.000

Table 4-5 Package Contents - Boot Loaders and Safe Loaders

Component	Version
Position Sensor Boot Loader	2019.03.05
Position Sensor Safe Loader	8.12
Video Camera Boot Loader	2018.01.17
Video Camera Safe Loader	8.12
SCU Boot Loader	2019.03.05
SCU Safe Loader	8.12

API Version

Table 4-6 Version Details - Package 008.012

API Revision - Position Sensor	G.003.006
API Revision - Video Camera Unit	V.001.000
API Revision - System Control Unit	G.003.006

New Commands

N/A

Modified Commands

The `INIT` and `RESET 0` commands now shut down all active streams.

The `BX2` command response has been modified for the new data averaging feature:

- Bit 9 in the frame status now indicates "Averaging Enabled".
- Code 5 in the tool status now indicates "Partially Averaged".

The `STREAM` command has been modified to no longer allow streaming of the following API commands:

- `INIT`
- `RESET`
- `STREAM`
- `USTREAM`

The `STREAM` command has also been modified to change the rate of streamed responses when not in tracking mode. Previously, the rate was fixed at 5 ms. Now, the rate is equivalent to the specified frame frequency.

Deprecated Commands

N/A

Removed Commands

N/A

New Parameters

Added `Info.Timeout.VGET` and `Info.Timeout.VSNAP` to the timeout table.

`Param.Clock.Time`: This parameter represents the system clock and is synchronized with the PTP Master clock. It can be used to measure the reply latency of `BX2` replies, using the embedded PTP time stamp, without requiring PTP support on the host. However, the accuracy is reduced.

Note **The integer part of the parameter represents seconds since epoch.**

The parameter `Param.Clock.Date Time` **represents the Vega's real-time clock (RTC). If the Vega Position Sensor is a PTP slave, the system clock and RTC will differ. The system clock is initialized by the RTC during start up but will be adjusted if an external PTP Master is found on the network.**

`Param.Data Averaging.Depth`: This parameter enables averaging of tracking data from the Position Sensor before it is reported to the host. Data averaging is performed on 3D positions (stray or tool 3D). At the end of the averaging period, 6D information is calculated on averaged 3D data. Averaging is performed with equal weights. For details, see the "*Polaris Vega Application Programming Interface Guide*".

`Param.Data Averaging.Threshold Distance`: This parameter specifies the maximum distance, in mm, that a measured 3D position can deviate from its estimated position without interrupting the marker's position history. For details, see the "*Polaris Vega Application Programming Interface Guide*".

Modified Parameters

Issuing a DFLT command no longer clears the `Info.Connect.isMaster` or `Info.Connections.isMaster` parameters.

`Param.Tracking.Frame Frequency` can now be set only in Setup mode. For details, see the "*Polaris Vega Application Programming Interface Guide*".

The maximum value for the `Param.Exposure.Shutter Time.Other` and `Cmd.VSnap.Manual Shutter` parameters is now set to 500 us for Vega XT.

Polling the `Info.Status.Gravity Vector` parameter no longer triggers an immediate reading of the accelerometer internally. Instead, it returns the latest cached value. The system internally reads the accelerometer between three and four times a second.

The visibility of some of the chirp parameters was modified. Previously, the duty cycle, span, and duration parameters could be accessed by users. Now, they are no longer accessible.

Deprecated Parameters

N/A

Removed Parameters

The following parameters have been removed:

- `Param.Connect.Host Port`
- `Param.Connect.Tracking Group`
- `Param.Network.Tracking Group`
- `Param.Network.DNS Servers`

New Features

Data Averaging

This release introduces an optional data averaging feature. Data averaging allows the user to trade data reporting frequency for a reduction in position jitter (noise). Data averaging is available for all Vega systems but is most useful on Vega XT because of the higher frame rate achievable.

Data averaging is disabled by default but can be enabled and configured via the following user parameters:

- `Param.Data Averaging.Depth`
- `Param.Data Averaging.Threshold Distance`

For details, see the "*Polaris Vega Application Programming Interface Guide*".

Increased Frame Rate and Lower Data Latency for Vega XT

This release introduces up to 400 Hz tracking frame rate and an average latency of less than 3 ms for Vega XT.

Active Wireless Tools for Vega XT

This release introduces support for active wireless tools for Vega XT.

Front Panel LED Behaviour

This release includes changes to the display on the front of Vega VT Position Sensors. When the video camera is streaming, the green front-panel LED will blink at 1 Hz with 80% duty cycle.

Note **Error and fault indications take precedence over the video streaming indication when they occur at the same time.**

UDP Streaming

This release introduces UDP streaming by extending the STREAM command to support options for specifying the destination port and protocol. For details, see the "*Polaris Vega Application Programming Interface Guide*".

Improvements and Modifications

Trailing Spaces are no Longer Accepted in API Commands

Firmware versions prior to CFR008 were more tolerant in how API commands were formatted. For example, in firmware prior to CFR008, the following was acceptable:

```
PINIT 01 \r   // whitespace between '01' and '\r'
```

Firmware after CFR008 no longer accepts this and will return "ERROR07" (Invalid number of parameters). Command syntax from firmware CFR008 and later is as follows:

```
PINIT 01\r   // (no space between '01' and '\r')
```

Child Connections (Streams) will not be Considered in Mastership Timeout

Ongoing activity on child connections generated from the parent connection do not reset the timeout counter of the parent connection. If there is no traffic on the parent connection for the time specified in `Param.Connect.Master Timeout`, mastership will be lost. By default, this is after 5 seconds. If no other connection issues a state-changing command, mastership will be regained immediately. For applications with a single parent connection, this has no effect.

If a stream is started on the parent connection itself, with no --port or --protocol parameter in the `STREAM` command, it will reset the mastership timer every time a `STREAM` reply is sent.

Note **It is the host's responsibility to keep the parent connection active. The host must ensure there is traffic on the parent connection within the interval specified by parameter `Param.Connect.Idle Timeout`. For details, see the "*Polaris Vega Application Programming Interface Guide*".**

USTREAM without any Options will Stop all Child Streams Associated with the Parent Connection

If more than one parent connection exists, only the child connections of the parent connection that issued the `USTREAM` command are stopped.

Improved Ethernet Status LED Indication

The status LEDs on the Ethernet connector now indicate the following information for the Position Sensor:

Table 4-7 LED 1 - Green - Ethernet Speed

Ethernet Speed	LED Indication
10 Mbps	single blink
100 Mbps	double blink
1000 Mbps	triple blink

Table 4-8 LED 2 - Amber - Link / Activity

Link / Activity	LED Indication
No Link	off
Link	on, solid
Activity	blink

Bug Fixes

Better Validation of List Parameters

Improved validation of list parameters to ensure invalid separator characters are not used.

GETINFO Responses

`GETINFO` responses were occasionally incomplete in previous releases.

Video Camera Frames Per Second

The video camera's frame rate per second when streaming is now automatically adjusted when switching video camera resolution.

For example, in previous releases, when switching from 2048 x 1536 (with a maximum frame rate of 20 Hz) to 1920 x 1088 (with a maximum frame rate of 30 Hz), the frame rate would stay at 20 Hz.

The video camera now automatically streams video at the maximum frame rate for the selected resolution.

Note **If the host manually set and saved the frame rate, the frames per second setting will persist (if within range) across all resolutions.**

LLDP Power Reporting

In previous releases, when Vega successfully completed two-event PHY classification, it did not update the LLDP record correctly and reported 13 W allocation. This did not affect the functionality of the system but was not in full compliance with the PoE standard.

Vega Position Sensors now correctly report 25.5 W power allocation.

INIT Command in Tracking Mode

Combined firmware 007 introduced an issue where, if an `INIT` command is issued while the Vega Position Sensor is in Tracking Mode, subsequent tracking sessions could start reporting a "Sensor Not Functioning" error and tracking could become unreliable. Package 008 corrects this issue.

Package 008 also fixes a second issue related to issuing an `INIT` command while in Tracking Mode where the value of `Param.Tracking.Track Frequency` would not be correctly restored to the value previously saved by the user.

Failed to Save

In previous versions, `Param.Exposure.Time Slot.Passive` failed to retain a saved value across a soft reset command (i.e. `RESET 0`).

PTP Time Change

The PTP Sync alert is no longer triggered erroneously when a PTP time change occurs.

In previous packages, when a PTP time change occurred, it would take 3 or 4 seconds for all Vega Position Sensors to start using the new master clock. For example, when a new PTP master was added to the subnet or the current master was removed. This was checked once per second and an alert was raised if an out-of-sync condition persisted for 3 seconds or longer.

To reduce the number of PTP Sync alerts, this time has been changed from 3 seconds to 10 seconds in Package 008.

STREAM Command Sequencing

Package 008 fixes an issue where a `STREAM` command issued before an `INIT` results in a stream of ERROR05 responses (time-out on command execution). Now, ERROR10 is returned and a stream is not set up.

False “Illuminator Current Out of Range” Faults

In previous releases, occasional false illuminator current faults were being raised. These faults were typically cleared automatically within 1-2 seconds, but issues could occur in some applications if the system was rebooted before the fault was cleared. Improvements in previous versions of firmware significantly reduced the frequency of these false faults. Package 008 includes changes that fully address this issue at its root cause.

Features.Volumes.* Doesn't Show Volumes

Package 008 fixes an issue where Features.Volumes.* may show no volumes although "SFLIST 03" shows the correct volumes.

Known Issues

Active Wired Tool Tracking

This firmware package does not support active wired tool tracking. Package 008.013 addresses this.

5 Vega Version 007 Firmware

This chapter provides a reference of the history of firmware revision 007 for the Polaris Vega System. It details the history of enhancements and fixes made to the Vega firmware.

5.1 Package 007.011

Package 007.011 introduces support for the Vega XT Position Sensor and includes a number of parameter changes, bug fixes, and improvements.

Package Contents

Table 5-1 Package Contents - Package 007.011 (February 2020)

Component	Version
Position Sensor File System	7.11
Position Sensor Firmware Compatibility File	007
Position Sensor Control Application	008.007.011
Position Sensor Driver Firmware	2.27.0
Position Sensor Main Board Logic for 7010	191205-113544
Video Camera File System	7.11
SCU File System	7.11
SCU Firmware Compatibility File	007
SCU Control Application	008.007.011
SCU Driver Firmware	2.27.0
SCU Main Board Logic	191205-114052
STB Control Firmware	000.002.005

Table 5-2 Package Contents - Boot Loaders and Safe Loaders

Component	Version
Position Sensor Boot Loader	2019.03.05
Position Sensor Safe Loader	7.11
Video Camera Boot Loader	2018.01.17
Video Camera Safe Loader	7.11
SCU Boot Loader	2019.03.05
SCU Safe Loader	7.11

API Version

Table 5-3 Version Details - Package 007.011

API Revision - Position Sensor	G.003.005
API Revision - Video Camera Unit	V.001.000
API Revision - System Control Unit	G.003.005

New Commands

N/A

Modified Commands

The INIT command has been modified to return WARNING06 instead of WARNING01 if issued from a Monitor connection when the system is in either Tracking or Diagnostics mode or has a tool loaded.

Deprecated Commands

N/A

Removed Commands

N/A

New Parameters

`Info.Status.Tracking.Measured Frame Frequency`: Reports the actual rate at which the system takes images of the measurement volume.

`Info.Status.Tracking.Measured Track Frequency`: Reports the actual rate at which the system reports transformations for all the tools being tracked.

Modified Parameters

`Param.Simulated Alerts:` This parameter value can now be saved.

`Param.Tracking.Frame Frequency:` This parameter value can be written and saved for Vega XT.

`Param.Tracking.Track Frequency:` This parameter value can now be saved.

`Param.Tracking.Selected Volume:` This parameter value can now be saved.

`Cmd.VSnap.Manual Shutter:` This parameter value can no longer be saved.

`Param.Exposure.Time Slot.Passive:` This parameter can no longer be set to a value that is greater than 1.

Deprecated Parameters

N/A

Removed Parameters

N/A

New Features

This version of firmware supports the Vega XT Position Sensor which offers a higher tracking rate than the Vega ST and VT Position Sensors. The tracking rate of the Vega XT Position Sensor is configurable up to 250 Hz. Compared to the MR250 option on the Vega ST Position Sensor, the Vega XT Position Sensor provides lower latency and better accuracy, meeting or exceeding the accuracy of the Vega ST and VT Position Sensors at 60 Hz.

Improvements and Modifications

Hosts Lists Ignored In Safe Mode

In previous versions of the firmware, setting `Param.Connect.Master Hosts` or `Param.Connect.Monitor Hosts` to an invalid value could result in the inability to connect to the Vega device. This version of firmware ignores these lists of hosts when in safe mode, thus allowing the user to correct the values of these parameters.

PTP Master Change Event

The Position Sensor API now indicates whenever the PTP master changes. This is done via a new event, “Event 6” in the system alert component of the BX2 response.

Default Chirp Duty Cycle

The default chirp duty cycle has been changed from 90% to 50%. This is beneficial for most scenarios because it provides increased trigger sensitivity in the IR receiver of active wireless tools.

Bug Fixes

Correct Use of Leap Seconds in PTP Slave Mode

When in PTP slave mode, the Vega Position Sensor was not subtracting the leap seconds value from the TAI time correctly to derive the UTC time. Thus, the system time, and the timestamp reported in the BX2 frame component, were 37 seconds (as of 2019) ahead of UTC time. This version of firmware corrects this problem.

False Insufficient PoE Power Alert

With multiple Vega Position Sensors on the same network it was possible in Package 004 that a false “PoE power supply insufficient for tracking” alert could be raised. The source of the problem is also present in firmware Package 006, however it does not exhibit itself to the user due to other firmware differences between Packages 004 and 006. This false alert is fixed in this version of firmware.

MR250 PTP Slave Crash While Tracking

With the MR250 option it was possible that a Vega Position Sensor could crash and reset while tracking if it was a PTP Slave and the PTP time changed abruptly by more than a couple of minutes. This version of firmware corrects this problem.

Transient Low Power PoE Alert

A transient low power PoE alert (0x40000) was set on the Vega Position Sensor at boot-up when it was connected to an LLDP network switch. The alert cleared very quickly and was only visible for a very short time in the `Info.Status.Alerts` parameter after a connection was possible, but was retained in the `Info.Status.New Alerts` parameter. If an INIT was attempted while the alert was set then an ERROR15 was returned. This version of firmware eliminates the transient alert.

New Alert Bit Not Reset

In Package 006, reading `Info.Status.New Alerts` would indicate that a simulated alert had been set via `Param.Simulated Alerts` however, the bit would remain set instead of resetting once read. This version of the firmware corrects this problem.

Past Dates Obstruct Publishing on mDNS

On Vega ST and VT Position Sensors running firmware Package 006, setting the hardware clock to a date before January 1, 2000 would cause the Position Sensor to not publish itself on mDNS. This version of the firmware corrects this problem.

PTP Time Jumps

For Vega systems running as PTP Master, discrete jumps in PTP time stamps of 8.59 seconds could occasionally be experienced, occurring, on average, about every 7.5 hours when not tracking and once every 15000 hours when tracking. Applications that make use of the PTP time stamp contained in the BX2 reply would see this jump, which could lead to a transient loss of synchronization between devices until the next PTP update is performed.

For Vega ST systems using serial communications, this bug could cause a spontaneous, automatic system reboot. This is because the Vega system FPGA runs a clock that is also synchronized to the PTP time and the loss of synchronization triggers a reboot as part of the auto-recovery strategy.

This version of firmware corrects this problem.

ST/VT Exposure Control

Previously, when one or more tools were loaded but none of the tools were in view, the exposure algorithm would set the exposure time to maximum and any stray markers that were not part of a tool were tracked with saturated signal. Package 007.011 fixes this issue. Now, the exposure algorithm correctly sets the exposure time so stray markers do not have saturated signals. However, the 3D position jitter for these stray markers might increase compared to previous versions of firmware.

Known Issues

N/A

6 Vega Version 006 Firmware

This chapter provides a reference of the history of firmware revision 006 for the Polaris Vega System. It details the history of enhancements and fixes made to the Vega firmware.

6.1 Package 006.023

Package 006.023 is a patch release built on-top of package 006.022.

Statement of Compliance

- This firmware patch does not alter the tracking algorithm.
- This firmware patch does not effect system accuracy.
- This firmware patch does not impact any regulatory classifications or approvals for Polaris Vega Position Sensors.

Package Contents

Table 6-1 Package Contents - Package 006.023 (June 2020)

Component	Version
Position Sensor File System	6.23
Position Sensor Firmware Compatibility File	006
Position Sensor Control Application	008.006.023
Position Sensor Driver Firmware	1.28.2
Position Sensor Main Board Logic for 7010	190830-174000
Video Camera File System	6.23
SCU File System	6.23
SCU Compatibility File	006
SCU Control Application	008.006.023
SCU Driver Firmware	1.28.2
SCU Main Board Logic	190829-094505
STB Control Firmware	000.002.005

Table 6-2 Package Contents - Boot Loaders and Safe Loaders

Component	Version
Position Sensor Boot Loader	2019.03.05
Position Sensor Safe Loader	6.23
Video Camera Boot Loader	2018.01.17
Video Camera Safe Loader	6.23
SCU Boot Loader	2019.03.05

Table 6-2 Package Contents - Boot Loaders and Safe Loaders

Component	Version
SCU Safe Loader	6.23

API Version

Table 6-3 Version Details - Package 006.023

API Revision - Position Sensor	G.003.004
API Revision - Video Camera Unit	V.001.000
API Revision - System Control Unit	G.003.002

New Commands

N/A

Modified Commands

N/A

Deprecated Commands

N/A

Removed Commands

N/A

New Parameters

N/A

Modified Parameters

N/A

Deprecated Parameters

N/A

Removed Parameters

N/A

New Features

N/A

Improvements and Modifications

N/A

Bug Fixes

Low Temperature Alert After Serial Break

This release corrects an issue where a spurious and transient low temperature alert could be raised immediately after a serial break.

Known Issues

N/A

6.2 Package 006.022

Package 006.022 is a patch release built on-top of package 006.021.

Statement of Compliance

- This firmware patch does not alter the tracking algorithm.
- This firmware patch does not effect system accuracy.
- This firmware patch does not impact any regulatory classifications or approvals for Polaris Vega Position Sensors.

Package Contents

Table 6-4 Package Contents - Package 006.022 (March 2020)

Component	Version
Position Sensor File System	6.22
Position Sensor Firmware Compatibility File	006
Position Sensor Control Application	008.006.022
Position Sensor Driver Firmware	1.28.2
Position Sensor Main Board Logic for 7010	190830-174000
Video Camera File System	6.22
SCU File System	6.22
SCU Compatibility File	006
SCU Control Application	008.006.022
SCU Driver Firmware	1.28.2
SCU Main Board Logic	190829-094505
STB Control Firmware	000.002.005

Table 6-5 Package Contents - Boot Loaders and Safe Loaders

Component	Version
Position Sensor Boot Loader	2019.03.05
Position Sensor Safe Loader	6.22
Video Camera Boot Loader	2018.01.17
Video Camera Safe Loader	6.22
SCU Boot Loader	2019.03.05
SCU Safe Loader	6.22

API Version

Table 6-6 Version Details - Package 006.022

API Revision - Position Sensor	G.003.004
API Revision - Video Camera Unit	V.001.000
API Revision - System Control Unit	G.003.002

New Commands

N/A

Modified Commands

N/A

Deprecated Commands

N/A

Removed Commands

N/A

New Parameters

N/A

Modified Parameters

`Param.Simulated Alerts`: This parameter can now be saved.

Deprecated Parameters

N/A

Removed Parameters

N/A

New Features

N/A

Improvements and Modifications

Vega VT Firmware Update

Firmware package 006.021 introduced instability into the firmware update process for Vega VT Systems, where up to half of attempted firmware updates could be unsuccessful. This version of firmware significantly reduces this instability.

Bug Fixes

PTP Time Jumps

For Vega systems running as PTP Master, discrete jumps in PTP time stamps of 8.59 seconds could occasionally be experienced, occurring, on average, about every 7.5 hours when not tracking and once every 15000 hours when tracking. Applications that make use of the PTP time stamp contained in the BX2 reply would see this jump, which could lead to a transient loss of synchronization between devices until the next PTP update is performed.

For Vega ST systems using serial communications, this bug could cause a spontaneous, automatic system reboot. This is because the Vega system FPGA runs a clock that is also synchronized to the PTP time and the loss of synchronization triggers a reboot as part of the auto-recovery strategy.

This version of firmware corrects this problem.

Known Issues

N/A

6.3 Package 006.021

Package 006.021 is a patch release built on-top of package 006.019 to address three specific issues.

Statement of Compliance

- This firmware patch does not alter the tracking algorithm.
- This firmware patch does not effect system accuracy.
- This firmware patch does not impact any regulatory classifications or approvals for Polaris Vega Position Sensors.

Package Contents

Table 6-7 Package Contents - Package 006.021 (September 2019)

Component	Version
Position Sensor File System	6.21
Position Sensor Firmware Compatibility File	006
Position Sensor Control Application	008.006.021
Position Sensor Driver Firmware	1.28.2
Position Sensor Main Board Logic for 7010	190830-174000
Video Camera File System	6.21
SCU File System	6.21
SCU Compatibility File	006
SCU Control Application	008.006.020
SCU Driver Firmware	1.28.2
SCU Main Board Logic	190829-094505
STB Control Firmware	000.002.005

Table 6-8 Package Contents - Boot Loaders and Safe Loaders

Component	Version
Position Sensor Boot Loader	2019.03.05
Position Sensor Safe Loader	6.21
Video Camera Boot Loader	2018.01.17
Video Camera Safe Loader	6.21
SCU Boot Loader	2019.03.05
SCU Safe Loader	6.21

API Version

Table 6-9 Version Details - Package 006.021

API Revision - Position Sensor	G.003.004
API Revision - Video Camera Unit	V.001.000
API Revision - System Control Unit	G.003.002

New Commands

N/A

Modified Commands

N/A

Deprecated Commands

N/A

Removed Commands

N/A

New Parameters

N/A

Modified Parameters

N/A

Deprecated Parameters

N/A

Removed Parameters

N/A

New Features

N/A

Improvements and Modifications

N/A

Bug Fixes

False “Temperature Sensor Not Functioning” Faults

This release corrects an issue where false “Temperature Sensor Not Functioning” faults were logged during system reboots. This was caused by an improper shut-down sequence that left the system in an indeterminate state. The shut-down sequence has been corrected to avoid this problem.

False “Illuminator Current Out of Range” Faults

In previous releases, occasional false illuminator current faults were being raised. These faults were typically cleared automatically within 1-2 seconds but issues could occur in some applications if the system was rebooted before the fault was cleared. To correct this issue and improve illuminator current fault handling, the following changes have been made:

- Corrected the timing of current and voltage readings from the hardware to ensure the data being analyzed is valid.
- Illuminator current and voltage readings are now analyzed for consistency before a fault is generated.

Video Camera Unit Safe Loader

The correct functionality of the Video Camera Unit (VCU) safe loader has been restored in package 006.021. In package 006.019, the Video Camera Unit incorrectly reboots when in safe mode.

Known Issues

N/A

6.4 Package 006.019

Package 006.019 is a significant release that introduces many new features, improvements, and bug fixes to the Vega firmware.

Package Contents

Table 6-10 Package Contents - Package 006.019 (June 2019)

Component	Version
Position Sensor File System	6.19
Position Sensor Firmware Compatibility File	006
Position Sensor Control Application	008.006.0019
Position Sensor Driver Firmware	1.28.0
Position Sensor Main Board Logic for 7010	190222-101021
Video Camera File System	6.19
System Control Unit File System	6.19
System Control Unit Compatibility File	006
System Control Unit Control Application	008.006.019
System Control Unit Driver Firmware	1.28.0
System Control Unit Main Board Logic	170202-094709
STB Control Firmware	000.002.005

Table 6-11 Package Contents - Boot Loaders and Safe Loaders

Component	Version
Position Sensor Boot Loader	2019.03.05
Position Sensor Safe Loader	6.19
Video Camera Boot Loader	2018.01.17
Video Camera Safe Loader	6.19
System Control Unit Boot Loader	2019.03.05
System Control Unit Safe Loader	6.19

API Version

Table 6-12 Version Details - Package 006.019

API Revision - Position Sensor	G.003.004
API Revision - Video Camera Unit	V.001.000
API Revision - System Control Unit	G.003.002

New Commands

The following commands are reintroduced in firmware version 006. For more information, see the "*Polaris Vega Application Programming Interface Guide*".

- PSEL
- PSRCH
- VGET
- VSNAP

Modified Commands

The VCAP command has been expanded to support the capture and return of images from the Vega video camera. For details, see the "*Polaris Vega Application Programming Interface Guide*".

The COMM command has always been supported in the firmware, but was not used by the Vega system. Vega systems equipped with a LEMO connector can use serial communications and therefore require use of the COMM command.

Deprecated Commands

N/A

Removed Commands

N/A

New Parameters

- `Param.Clock.Date Time`: Used to set and retrieve the system time. All other Clock parameters are deprecated. See "Deprecated Parameters" on page 36.
- `Features.Firmware.API Revision`: The recommended way to retrieve the Vega API revision information. The APIREV command is still supported.
- `Param.Serial Break Option`: Used for serial communications, to configure the desired action upon a serial break.
- There are many new parameters for controlling the video camera configuration. For details, see the "*Polaris Vega Application Programming Interface Guide*".

Modified Parameters

The following parameters may now only be set in Setup mode. ERROR0C will be returned in Tracking and Diagnostic modes:

- `Param.Tracking.Selected Volume`
- `Param.Tracking.Track Frequency`
- `Param.Tracking.Sensitivity.Active`
- `Param.Tracking.Sensitivity.Active Wireless`
- `Param.Tracking.Sensitivity.Passive`

Deprecated Parameters

The following deprecated parameters are replaced with the new `Param.Clock.Date Time` parameter:

- `Param.Clock.Date`
- `Param.Clock.Param.Clock.Day of Week`
- `Param.Clock.Hours`
- `Param.Clock.Minutes`
- `Param.Clock.Year`
- `Param.Clock.Month`

In package 006 these parameters are no longer visible by default, but can still be accessed if the full parameter name is used.

Removed Parameters

N/A

New Features

Support for Video Camera

This version of the firmware supports the optional Vega video camera. The Vega video camera provides a live, colour video stream of the Vega measurement volume. The video camera is not used for tracking tools but it is possible to project the Position Sensor's tool tracking data stream onto the live video stream. This feature can be used for augmented reality overlays. See the user guide shipped with your system for details.

Support for Serial Communications

This version of the firmware supports serial communications between the Vega Position Sensor and host computer. Only Vega Position Sensors purchased with the LEMO connector option support this feature. Video camera functionality is not supported via serial communications. This option is only recommended for customers that are migrating from Polaris Spectra to Polaris Vega. It is not recommended for new designs. See the user guide shipped with your system for details.

Radiation Robustness

This version of the firmware supports a radiation-hardened variant of the Vega Position Sensor. This variant is suited for use in radiation oncology environments where there is potential exposure to gamma and neutron radiation. This feature enables modified memory management.

Improvements and Modifications

Boot Time Reduction

A number of changes were made to decrease the boot time of the Vega System.

Better Radix Tool Tracking

The exposure control algorithm has been adjusted to improve Radix tool tracking stability.

Improved Command Response Times

Response times for the INIT and TSTART commands are now more consistent and shorter in duration.

Improved Error Handling

Error detection in the firmware (and FPGA) has been improved to minimize the possibility of lock-up conditions, preferring auto-recovery instead, as facilitated by an application or device reboot. This enhanced robustness is especially applicable for neutron radiation environments. In the event an auto-reboot occurs, associated information is entered into the Event Log.

Improved Event Logging

More detailed information is recorded to the Event Log after firmware updates.

Improvements to Parameters

Changes to the `Param.Tracking.Sensitivity.*` parameters now take effect immediately rather than when the exposure time next changes.

The values reported by the parameter `Info.Status.Gravity Vector` have been reduced to two decimal places to match the precision of the data.

The System Control Unit is no longer registered in the Position Sensor’s device table. Therefore, the System Control Unit's parameter system is no longer proxied through the Position Sensor, i.e., these parameters are no longer accessible via the Position Sensor by using the SCU-0. parameter name prefix.

Change to PTP Master Clock Offset

When a Vega Position Sensor becomes PTP Master, it now publishes a UTC Offset of 0 sec instead of 35 sec. PTP time is reported in TAI format (i.e. not adjusted for leap seconds). Vega's internal clock is initially set to UTC during manufacturing, but will drift away from accurate UTC time over time. This means the absolute time value should not be used. The main purpose of Vega being the

PTP master is that a common time base is established between all PTP devices. If the correctness of absolute time is important, NDI recommends using a PTP Grandmaster clock in the system.

Improved Fault and Alert Handling

Reporting of individual System Control Unit fault and alert codes has been added as part of the BX2 system alerts component.

The alert bit 0x00010000 was introduced to indicate when the firmware is executing in Safe Mode.

The alert bit 0x00020000 was introduced on the Position Sensor to indicate when the System Control Unit is in an error state.

Handling of the simulated bump sensor alert is improved to more closely resemble the actual hardware bump sensor alert.

When a bump is detected, the axis in which the bump occurred is no longer detected or reported.

Change to Master Mode Connection Timeout

The master mode connection timeout is now based on last response time, not time of last received command.This means that if a master connection is set up for streaming, mastership stays with this connection, even if the host side does not issue any commands.

Bug Fixes

Event Log Fixes

Package 006 fixes the Event Log entry for simulated bumps so it shows the current system time rather than an invalid time value latched in the hardware real-time clock.

Persistent alerts are no longer duplicated in the Event Log.

Deadlock Situation After Unblocking Sensor Fixed

In packages 003 and 004, under specific conditions when tracking a static scene, temporarily blocking and then unblocking one of the IR sensors would result in an error status of “Too Few Markers” being returned for an indeterministic amount of time. This situation was caused by the exposure control algorithm keeping the image in saturation, which in turn prevented the tool from tracking. Package 006 addresses this by ensuring that tracking immediately resumes in this scenario.

PENA Command Fixed

PENA no longer calls PINIT if the tool is already initialized.

Program Crashes Fixed

Fixed program crash due to providing too many arguments to an API command.

Fixed program crash that would occasionally occur when a GETINFO Features.* command was executed after an INIT or RESET 0 command.

Fixed a program crash that would occur occasionally when issuing the PINIT command.

Real-Time Clock Fixed

In package 006, the system real-time clock can be set when the internal battery is fully drained.

System Control Unit Related Fixes

Package 006 no longer raises a spurious System Control Unit firmware incompatibility error upon INIT.

The System Control Unit no longer retains connections to the Position Sensor after the Position Sensor power cycles.

Known Issues

N/A

7 Vega Version 004 Firmware

This chapter provides a reference of the history of firmware revision 004 for the Polaris Vega System. It details the history of enhancements and fixes made to the Vega firmware.

7.1 Package 004.008

This release introduces support for the MR250 tracking rate along with a number of improvements and bug fixes.

Package Contents

Table 7-1 Package Contents - Package 004.008 (June 2018)

Component	Version
Position Sensor File System	3.8
Position Sensor Firmware Compatibility File	004
Position Sensor Control Application	008.004.007
Position Sensor Driver Firmware	1.16.0
Position Sensor Main Board Logic for 7010	180108-115832
SCU File System	3.8
SCU Compatibility File	004
SCU Control Application	008.004.007
SCU Driver Firmware	1.16.0
SCU Main Board Logic	170202-094709
STB Control Firmware	000.002.005

Table 7-2 Package Contents - Boot Loaders and Safe Loaders

Component	Version
Position Sensor Boot Loader	2017.12.19
Position Sensor Safe Loader	3.8
SCU Boot Loader	2017.12.19
SCU Safe Loader	3.8

API Version

Table 7-3 Package 004.008

API Revision - Position Sensor	G.003.002
API Revision - System Control Unit	G.003.002

Support for Video Camera Removed

This version of the firmware does not support the Vega Video Camera.

New Commands

N/A

Modified Commands

N/A

Deprecated Commands

N/A

Removed Commands

N/A

New Parameters

N/A

Modified Parameters

N/A

Deprecated Parameters

N/A

Removed Parameters

N/A

New Features

Support for 250 Hz at Quarter Sensor Resolution

Package 004 of the Polaris Vega firmware supports the use of the MR250 feature key, which allows for tracking at a frame rate of 250Hz. The MR250 feature key is an additional option offered by NDI and is not, by default, included in Package 004. An upgrade of an existing Vega system to Package 004 will not include the MR250 feature key unless it is explicitly provided by NDI.

In the 250Hz tracking mode, the IR sensors are set to quarter resolution. As a result, there are limitations on the types of tools that can be tracked, as well as a reduced accuracy specification. The user guide for your system includes detailed performance specifications for this feature.

Note **For instructions on enabling and disabling feature keys, see the *"Polaris Vega Application Programming Interface Guide"* or the *"NDI ToolBox online help"*.**

Improvements and Modifications

Support for IEEE 802.3at LLDP

Package 004 introduces data link layer classification (LLDP) as mandated by the IEEE 802.3at standard. This ensures that the Vega Position Sensor can be successfully connected to and powered by any 802.3at compliant switch.

Improved Passive Tool Tracking in Presence of Many Stray Markers

In cases where passive tools are tracked in the presence of an excess of stray markers (markers not associated with tools), there is a chance that pairs of stray markers match segments on the tool. The unique geometry algorithm uses both the segment length and the angle between pairs of segments to determine which of the measured 3D positions belong to the tool. If, from the set of stray markers, two segment pairs happen to match both in length and angle with a pair of tool segments, they can be incorrectly associated with the tool. This incorrect association will be detected during the subsequent rigid body fitting algorithm and therefore no incorrect pose transformation will be reported. However, it is possible that the system will end up in a situation where the tool cannot "lock-on", and therefore no pose information will be obtained for the tool even though it is in view. Package 004 reduces the likelihood of this happening by adding a requirement that the measured segment pair must share a common vertex marker.

GET and SET Commands on Position Sensor Accept "PS-0" Prefix

To provide backwards compatibility with Polaris Spectra, Package 004 supports GET and SET commands that specify a "PS-0" parameter name prefix.

Bug Fixes

Illuminator Current Monitoring

The thresholds used for illuminator current monitoring were incorrect and too tight. This could result in spurious illuminator current alerts. Package 004 fixes this by setting the correct limits.

Frame Counter Starts at Incorrect Number

In rare cases, due to internal image synchronization methodology, previous firmware starts the frame number at an incorrect value (typically 1). Package 004 fixes this, ensuring the frame number always starts at the correct (PTP-derived) value.

Bump Event Timestamp

In Package 004, the timestamp recorded to the Position Sensor Event Log is the actual time at which the bump event occurred (made possible by the battery-backed Real Time Clock), not the time at which the Position Sensor powers up and notices the bump event.

Duplicate STREAM IDs

Package 004 no longer starts another stream when the same stream ID is specified. Instead, the existing stream is replaced by the new stream.

STREAM --repeat Option

Package 004 firmware now correctly honours the --repeat option of the STREAM command.

Incorrect Param.Clock.* Parameter Values

The components of time (year, month, day, hour, minute, second) are represented in the `Param.Clock.*` parameters. When setting one of these parameters, unless it had been read immediately before, the rest of the component parameters would be set to incorrect values. These parameters could also end up with incorrect values when a “RESET 1” command is issued. Package 004 fixes both problems.

Missing Active Wired Tool Upon Introduction of New PTP Master

Introduction of a new PTP master could cause the SCU strober to lose synchronization, thereby causing active wired tools to go missing indefinitely. The only way to recover would be to stop and start tracking. Package 004 fixes this problem by automatically recovering synchronization, though active wired tools may still go missing for a short period of time (i.e. seconds) during the recovery process. Note that introduction of a new PTP master into a network while in tracking mode is undesirable and should be avoided.

PTP Sync Alerts

Sporadic PTP sync alerts have been observed, even in point-to-point connected networks where there is no second PTP device. The problem has been traced to inadequate robustness in Vega firmware related to PTP status handling. This has been addressed in Package 004.

Known Issues

The following issues relate to the Vega product shipped with firmware version 004.

Marker Wavelength

Customers who use active wired tools with ROM files that do not define the marker wavelength should note that for such tools, the Polaris Vega System assumes that the marker wavelength is 880 nm. The previous version of the system (the Polaris Spectra) assumed that active wired tools were built with 930 nm markers if the wavelength was not specified in the ROM file. The Polaris Vega only supports 880 nm markers. Users need to ensure that they use only NDI 880 nm active markers with Vega. Failure to do so may result in inaccurate data being returned by the system.

8 Vega Version 003 Firmware

This chapter provides a reference of the history of firmware revision 003 for the Vega System. It details the history of enhancements and fixes made to the Vega firmware.

8.1 Package 003.017

Package 003.017 is a patch release built on-top of package 003.016 to address two specific issues.

Statement of Compliance

- This firmware patch does not alter the tracking algorithm.
- This firmware patch does not effect system accuracy.
- This firmware patch does not impact any regulatory classifications or approvals for Polaris Vega Position Sensors.

Package Contents

Table 8-1 Package Contents - Package 003.017 (September 2019)

Component	Version
Position Sensor File System	2.8
Position Sensor Firmware Compatibility File	003
Position Sensor Control Application	008.003.017
Position Sensor Driver Firmware	1.12.1
Position Sensor Main Board Logic for 7010	190829-094109
SCU File System	2.8
SCU Compatibility File	003
SCU Control Application	008.003.015
SCU Driver Firmware	1.12.1
SCU Main Board Logic	190829-094505
STB Control Firmware	000.002.005

Table 8-2 Package Contents - Boot Loaders and Safe Loaders

Component	Version
Position Sensor Boot Loader	2017.5.30
Position Sensor Safe Loader	003
SCU Boot Loader	2017.5.30
SCU Safe Loader	003

API Version

Table 8-3 Version Details - Package 003.017

API Revision - Position Sensor	G.003.002
API Revision - System Control Unit	G.003.002

New Commands

N/A

Modified Commands

N/A

Deprecated Commands

N/A

Removed Commands

N/A

New Parameters

N/A

Modified Parameters

N/A

Deprecated Parameters

N/A

Removed Parameters

N/A

New Features

N/A

Improvements and Modifications

N/A

Bug Fixes

False “Temperature Sensor Not Functioning” Faults

This release corrects an issue where false “Temperature Sensor Not Functioning” faults were logged during system reboots. This was caused by an improper shut-down sequence that left the system in an indeterminate state. The shut-down sequence has been corrected to avoid this problem.

False “Illuminator Current Out of Range” Faults

In previous releases, occasional false illuminator current faults were being raised. These faults were typically cleared automatically within 1-2 seconds but issues could occur in some applications if the system was rebooted before the fault was cleared. To correct this issue and improve illuminator current fault handling, the following changes have been made:

- Thresholds for both illuminator current and voltage monitoring have been adjusted to align with the physical hardware limits more accurately.
- Corrected the timing of current and voltage readings from the hardware to ensure the data being analyzed is valid.
- Illuminator current and voltage readings are now analyzed for consistency before a fault is generated.
- When an illuminator-related fault is generated, more information (e.g. illuminator serial number, current or voltage value) is now written to the Event Log.

Known Issues

N/A

8.2 Package 003.016

Package 003.016 is a patch release built on-top of package 003.015 to address one issue.

Statement of Compliance

- This firmware patch does not alter the tracking algorithm.
- This firmware patch does not effect system accuracy.
- This firmware patch does not impact any regulatory classifications or approvals for Polaris Vega Position Sensors.

Package Contents

Table 8-4 Package Contents - Package 003.016 (June 2018)

Component	Version
Position Sensor File System	2.7
Position Sensor Firmware Compatibility File	003
Position Sensor Control Application	008.003.015
Position Sensor Driver Firmware	1.12.0
Position Sensor Main Board Logic for 7010	170605-140801
SCU File System	2.7
SCU Firmware Compatibility File	003
SCU Control Application	008.003.015
SCU Driver Firmware	1.12.0
SCU Main Board Logic	170202-094709
STB Control Firmware	000.002.005

Table 8-5 Package Contents - Boot Loaders and Safe Loaders

Component	Version
Position Sensor Boot Loader	2017.5.30
Position Sensor Safe Loader	003
SCU Boot Loader	2017.5.30
SCU Safe Loader	003

API Version

Table 8-6 Version Details - Package 003.016

API Revision - Position Sensor	G.003.002
API Revision - System Control Unit	G.003.002

New Commands

N/A

Modified Commands

N/A

Deprecated Commands

N/A

Removed Commands

N/A

New Parameters

N/A

Modified Parameters

N/A

Deprecated Parameters

N/A

Removed Parameters

N/A

New Features

N/A

Improvements and Modifications

Improved Network Security

Package 003.016 provides enhanced network security. Package 003.016 with File System version 2.7 supersedes Package 003.015 with File System version 2.6. To confirm that Package 003.016 is running on your Vega Position Sensor, ensure that `Features.Firmware.Package Number` returns “003.016”.

Bug Fixes

N/A

Known Issues

N/A

8.3 Package 003.015

Package 003.015 is a significant release that introduces changes to the Polaris Vega API, new features, and several important improvements and modifications.

Package Contents

Table 8-7 Package Contents - Package 003.015 (September 2017)

Component	Version
Position Sensor File System	2.6
Position Sensor Firmware Compatibility File	003
Position Sensor Control Application	008.003.015
Position Sensor Driver Firmware	1.12.0
Position Sensor Main Board Logic for 7010	170605-140801
Video Camera File System	3.15
SCU File System	2.6
SCU Firmware Compatibility File	003
SCU Control Application	008.003.015
SCU Driver Firmware	1.12.0
SCU Main Board Logic	170202-094709
STB Control Firmware	000.002.005

Table 8-8 Package Contents - Boot Loaders and Safe Loaders

Component	Version
Position Sensor Boot Loader	2017.5.30
Position Sensor Safe Loader	003
SCU Boot Loader	2017.5.30
SCU Safe Loader	003

API Version

Table 8-9 Version Details - Package 003.015

API Revision - Position Sensor	G.003.002
API Revision - System Control Unit	G.003.002

Configuration Change Notification

The API now indicates when the system configuration has changed. This happens when, for example, an associated SCU is connected or disconnected during tracking. For the BX and TX commands, this is done via “Bit 10” of the system status. For BX2 this is done via “Event 5” in the system alert component.

New Commands

The following commands are new to firmware version 003. For more information, see the "*Polaris Vega Application Programming Interface Guide"*.

- PFSEL

Modified Commands

Package 002 has various inconsistencies in the API that are different from the behaviour of Polaris Spectra, or diverge from the documentation. The table below compares the differences between API Revisions G.003.001 and G.003.002.

Table 8-10 API Modifications in Package 003.015

Bug Description	API G.003.001	API G.003.002
Incorrect error code when trying to enable a port handle that has become unoccupied	Reply to PENA → ERROR2B	Reply to PENA → ERROR2C
When requesting a dummy tool via PRHQ, the port is automatically enabled. This is inconsistent with standard behaviour	Port handle for dummy tool is automatically enabled (no PENA required)	Port handle for dummy tool is not automatically enabled (PENA required)
INIT command succeeds even though a "Recoverable parameter fault" condition is detected	Reply to INIT→ OKAY	Reply to INIT → ERROR15 (unable to initialize)
VCAP times out when issued on an empty sequence (no port handles enabled)	Reply to VCAP → ERROR05 (timeout)	Reply to VCAP → ERROR0F (port handle not enabled)
VCAP times out when issued with arguments "--frame=illuminated" or "--frame=background" if these have not been added to the sequence beforehand	Reply to VCAP → ERROR05 (timeout)	Reply to VCAP → ERROR23 (command parameter out of range)
DSTOP returns OKAY when issued in tracking mode	TSTART (tracking mode) Reply to DSTOP → OKAY	TSTART (tracking mode) Reply to DSTOP → ERROR0C (command invalid in current mode)
TSTOP returns OKAY when issued in diagnostics mode	DSTART (diagnostics mode) Reply to TSTOP → OKAY	DSTART (diagnostics mode) Reply to TSTOP → ERROR0C (command invalid in current mode)
Incorrect error message in BX2 when options are specified without value, for example BX2 --3d	Reply to BX2 –3d → ERROR14 (invalid position sensor characterization)	Reply to BX2 –3d → ERROR23 (command parameter out of range)
PHRQ accepts random characters in "Hardware Device" field, when only wild-card characters "*" should be accepted	Reply to PHRQ abcdefgh*1**** → OKAY	Reply to PHRQ abcdefgh*1**** → ERROR23 (command parameter out of range)
Information reported by Features.Tools.* does not match information reported by SFLIST	Features.Tools.Active Ports is not updated when SCU is connected/disconnected.	Features.Tools.* parameters now match information reported by SFLIST command. Exception: SFLIST never reports more than 15 (0xf) ports, because format is limited to 1 digit.
BX/TX allow invalid parameter options	Reply to TX 0021 → same as TX 0001 (valid response)	Reply to TX 0021 → ERROR23 (command parameter out of range)

The following commands or command replies have been modified to provide more accurate information.

Table 8-11 Modified commands and Replies

Description	API G.003.001	API G.003.002
BX/TX: Added notification for system configuration change, for example, SCU connected/disconnected	N/A	Added System Status, Bit 10: Configuration Change
BX2: Added notification for system configuration change, for example SCU connected/disconnected	N/A	Added Event 5 (Configuration Change) to System Alert Component

Deprecated Commands

N/A

Removed Commands

N/A

New Parameters

- `Param.Video Camera.Allow Streaming`: Enables and disables video camera streaming.
- `VCU-0.*`: Parameters for accessing the video camera configuration. For details, see the "*Polaris Vega Application Programming Interface Guide"*.
- `Features.Firmware.Bootloader Version`: Displays the version of the boot loader.

Modified Parameters

Table 8-12 Modified Parameters in Package 003.015

Parameter	API G.003.001	API G.003.002
Info.Status.Alerts, bit 0x10000000	Storage Temperature Exceeded	Reserved

Deprecated Parameters

N/A

Removed Parameters

Storage Temperature Detector Functionality Removed Package 003 no longer supports the extreme storage temperature detector. Parameters `Param.Temperature Detector.*` and `Info.Status.Temperature Exceeded` are no longer available. The system will no longer set the `Storage Temperature Exceeded` flag (bit 0x10000000) in the `Info.Status.Alerts` parameter. To ensure sufficient accuracy after shipment, NDI recommends that the user performs an accuracy assessment at the install-site before using the system in critical applications. This can be done with the NDI Accuracy Assessment Kit or a user defined protocol.

New Features

Video Camera Unit

Package 003 now supports an optional Video Camera Unit (VCU). The Video Camera Unit delivers compressed live video using the RTSP protocol. Standard video clients like VLC and GStreamer can be used to receive the video stream. See the *"Polaris Vega User Guide"* for details.

Default Value for Marker Angle

If the MarkerAngle parameter is not specified in an SROM, a 60° default is used.

Notification if System is Connected to an Underpowered PoE Supply

Polaris Vega requires a 802.3at-compliant (Type 2) power supply. Firmware Package 002 does not check for this, and when connected to a PoE that is not 802.3at-compliant, it is possible for the system to experience brown-outs when tracking passive tools.

Package 003 checks the power available from the supply and if it is less than the minimum power required by the system, the alert "System Underpowered" (bit 18 in alerts) is raised. In addition, an attempt to initialize the system will return an error.

Improvements and Modifications

Improved Tracking Performance in Presence of Interference

In the presence of interference signals, Package 002 could relatively easily be prevented from tracking. Possible causes of interference are:

- Presence of direct or reflected sunlight or other light sources with high spectral energy content in the NIR range (800-1100nm). Depending on the situation, these light sources can be significantly brighter than the signal from the markers.
- Specular reflective surfaces like mirrors, polished metal or glass panes that reflect the cameras illuminator light (when tracking passive tools). Similar to sunlight, these reflections can be brighter than the signal from the markers.
- Diffuse reflective surfaces that appear with enough intensity to generate areas above the trigger threshold. The actual light-source can be external (sunlight etc.) or internal (light coming from system illuminators). In the sensor image, these areas are typically of lower intensity than the marker signal. Depending on the situation, the increased processing load may cause the system to be unable to complete processing the image in the allocated time.

Several modifications to the firmware were implemented to improve overall tracking performance in the presence of interference. These are:

Sensitivity Levels: Changed Trigger Threshold from Constant to Linear with Exposure Time Package 002 used constant trigger thresholds, which were independent of exposure time. This can cause tools to go missing if the signal is reduced below the trigger threshold because of the presence of high-intensity external IR light. Package 003 uses trigger thresholds that are linear with exposure time. If a bright signal causes a reduction in exposure time the trigger threshold will be lowered proportionally, making it possible to detect markers even if the signal is reduced.

Enhanced Exposure Control Algorithm Package 002 uses the brightest object in the scene as the input to exposure control. In the presence of high-intensity external IR (for example sunlight) this can lead to undesirable behaviour where tools go missing. Package 003 uses an adaptive algorithm that tries to consider only signals from valid tool markers if SROMs are loaded. In the case of stray marker tracking (i.e. only dummy tools loaded), the exposure control still tries to only consider signals that belong to valid 3D positions. However, it should be noted that under certain conditions, external light sources that are visible by both sensors may appear as valid 3D positions. This may cause undesirable behaviour of exposure control in rare situations. Image captures will identify the problem in such situations.

Improved Performance in 2D Processing In addition to the changes to sensitivity levels and exposure control, improvements to the 2D processing were implemented. These improvements reduce the processing time in “busy scenes” (scenes with high IR content) significantly. In comparison to Package 002, Package 003 can now handle much more complex scenes. In addition, if the system is not able to process all data in the allotted time, it will report this by indicating a “Fell Behind” condition in the port/tool status field, whereas Package 002 silently gave up.

Monotonic Frame Numbers Guaranteed During a Tracking Session

Package 002 used a fixed relationship between PTP time and the frame number reported. This could lead to situations where, in absence of a PTP grandmaster clock, if another device with PTP enabled was plugged into the network, PTP would re-negotiate the master. If the new device happened to become master and its local clock was offset from the previous PTP master, the new master’s clock would be used for the PTP time base. In this case, the frame numbers in Package 002 would experience a discrete jump of arbitrary size in either direction, e.g. the frame number could become smaller between subsequent frames.

Package 003 avoids these jumps by calculating a PTP-based frame number only at the beginning of tracking (TSTART command) and from then on relies on counting video frames. This means frame numbers will increment by one every $1/60^{th}$ of a second, based on a 60fps frame rate. If the PTP master changes during a tracking session, the frame number will not be impacted, even though the PTP time stamp will be. Note that at the beginning of the next tracking session, i.e. after a TSTOP/TSTART transition, the frame number will be re-synchronized with PTP. This means that monotonic frame numbers are not guaranteed between subsequent tracking sessions.

Bug Fixes

Improvements to Internal Process Management to Avoid Missing Frames

Package 002 contains code where dynamic process creation can cause the system to be slow to respond to tracking requests. If this happens the client can get a delayed response to a tracking request and miss several frames of tracking data.

Package 003 removes this inefficiency and immediately responds to tracking requests, thereby avoiding skipped frames.

Simulated Alerts Treated as True Alerts After Reset 1

Package 002 accidentally converted a simulated alert into a real alert after RESET 1. This bug is fixed in Package 003

Low Temperature Alert Bit Accidentally sets Diagnostic Pending Bit

In Package 002, the “Diagnostic Pending” bit (bit 8 in TX/BX System Status) was set in addition to “Operating Temperature Out-of-Range” (bit 9 in TX/BX System Status) during system warm up. After the system reached operating temperature, the “Operating Temperature Out-of-Range” flag was cleared automatically, but “Diagnostic Pending” flag stayed set. This caused BX and TX to not report transformations unless option 0x0800 was used, or the user cleared the “Diagnostic Pending” flag by reading the `Info.Status.New Alerts` parameter. This bug is fixed in Package 003. The “Diagnostic Pending” flag is no longer set if the temperature is low during warm up and as a consequence, TX/BX immediately start reporting transformations when the system has reached operating temperature conditions, without the need for using option 0x0800 or reading the `Info.Status.New Alerts` parameter.

Reporting of Stale Transformation Data

In Package 002, it was, in rare occasions, possible that stale tool transformation data was reported. In particular, this has been observed by tracking a passive tool and blocking one of the sensors from viewing the tool (it is, however, not easy to reproduce the behaviour consistently). In this case, the system kept reporting the exact last transformation. Package 003 fixes this.

Bump-Alert set even when Bump Detector is Disabled

In Package 002, when the bump sensor is disabled via parameter `Param.Bump Detector.Bump Detection=0`, a subsequent bump still set the bump detected flag (0x00200000) in the `Info.Status.Alerts` parameter. Package 003 no longer sets the bump detected flag when bump detection is disabled.

Known Issues

The following issues relate to the Vega product shipped with firmware version 003.

Marker Wavelength

Customers who use active wired tools with ROM files that do not define the marker wavelength should note that for such tools, the Polaris Vega System assumes that the marker wavelength is 880 nm. The previous version of the system (the Polaris Spectra) assumed that active wired tools were built with 930 nm markers if the wavelength was not specified in the ROM file. The Polaris Vega only supports 880 nm markers. Users need to ensure that they use only NDI 880 nm active markers with Vega. Failure to do so may result in inaccurate data being returned by the system.

Video Camera Disconnection

In a Vega system equipped with a video camera, the video stream is delivered via an RTSP connection, which supports delivery using UDP or TCP packets. Packet type will be determined by the video client that is running on the host computer. UDP packets should be used to deliver the video data. Failure to use UDP packets may result in high latency and possible video camera crashes.

Polaris Vega Occasionally Reports the PTP-Sync Error Incorrectly

This problem only occurs in network configurations with multiple PTP-enabled devices.

If the Polaris Vega is a PTP-Slave, then the system may occasionally report that the PTP-Clock is not synchronized (parameter `Info.Status.Alerts`, bit 0x08000000). Tests at NDI have shown that these incidents are “false positives”, i.e. the underlying PTP synchronization is still working and the error indication is incorrect. This error is transient and typically corrects itself within a few seconds.

NDI recommends the following work-arounds:

- Client applications that use Polaris Vega in a stand-alone mode, i.e. a single Polaris Vega Position Sensor tracking either passive or active wireless tools, should ignore the “PTP-Clock not Synchronized” alert.
- Client applications that use Polaris Vega in a configuration that requires PTP synchronization, i.e. either active wired tracking or multiple synchronized Polaris Vega systems should implement a timeout for the “PTP-Clock not synced” alert. If the alert is transient, the client should not react. The client should only react if the “PTP-Clock not synced” alert persists for longer than the specified timeout value. NDI recommends a default timeout value of 3 seconds.

Polaris Vega and Polaris Vega SCU May Lose Synchronization if the PTP-Master Changes During Tracking

This problem only affects systems configured for wired active marker tracking, i.e. systems including a Polaris Vega SCU.

PTP synchronization is used to synchronize the timing between the Polaris Vega Position Sensor and the Polaris Vega SCU. During the tracking of active markers, if the PTP master on the network changes (for example, by plugging another PTP enabled device into the network), it is possible that synchronization between the Polaris Vega Position Sensor and the Polaris Vega SCU is lost and active marker tracking is interrupted. When this happens, the system does not recover automatically. Re-starting tracking (by initiating a TSTOP/TSTART transition) will bring the system back into synchronization and tracking will resume properly.

NDI recommends the following work-arounds to prevent this issue:

- Use a dedicated subnet for the Polaris Vega and Polaris Vega SCU.
- Avoid adding/removing PTP enabled devices to the network during tracking sessions.

Video Camera White Balance Factory Presets Can Be Overwritten

NDI ships each video camera with one of two different lens types, standard or premium. The white balance presets for the predefined lighting conditions are slightly different for the two lens types. These values are stored as user parameters and can be overwritten by the user. In addition, if the user issues the DFLT * command followed by a SAVE command, the factory presets will be overwritten with hard-coded defaults.

NDI provides parameter files on the support site that can be uploaded to the camera to restore the factory presets. Gain access to the support site using this link: https://support.ndigital.com. To retrieve the correct parameter file, you need to know the model number for the video camera’s optical lens. Use the VCU-0.Features.Hardware.Model parameter to determine the model number.

9 Vega Version 002 Firmware

9.1 Package 002.005

This package introduces important new commands for use with Polaris Vega streaming data and removes and deprecates several 002 commands.

Package Contents

Table 9-1 Package Contents - Package 002.005 (August 2016)

Component	Version
Position Sensor File System	1.3
Position Sensor Firmware Compatibility File	002
Position Sensor Control Application	008.002.061
Position Sensor Driver Firmware	1.10.3
Position Sensor Main Board Logic for 7010	160809-163006
SCU File System	1.3
SCU Compatibility File	002
SCU Control Application	008.002.060
SCU Driver Firmware	1.10.3
SCU Main Board Logic	160614-142909
STB Control Firmware	000.002.005

Table 9-2 Package Contents - Boot Loaders and Safe Loaders

Component	Version
Position Sensor Boot Loader	2016.8.24
Position Sensor Safe Loader	002
SCU Boot Loader	2016.8.24
SCU Safe Loader	002

API Version

Table 9-3 Version Details - Package 002.005

API Revision - Position Sensor	G.003.001
API Revision - System Control Unit	G.003.001

New Commands

The following commands are new to firmware version 002. For details, see the "*Polaris Vega Application Programming Interface Guide*".

- BX2
- STREAM
- USTREAM
- VCAP

Modified Commands

N/A

Deprecated Commands

The following commands are deprecated. For details, see the "*Polaris Vega Application Programming Interface Guide*".

- 3D
- IRATE
- PINIT
- SFLIST
- VSEL

Removed Commands

The following commands are removed. For more information, see the "*Polaris Vega Application Programming Interface Guide*".

- GETIO
- HCWDOG
- PSEL
- PFSEL
- PSOUT
- PSRCH
- SENSEL
- SETIO
- SSTAT
- VGET
- VSNAP

New Parameters

N/A

Modified Parameters

N/A

Deprecated Parameters

N/A

Removed Parameters

N/A

New Features

N/A

Improvements and Modifications

N/A

Bug Fixes

N/A

Known Issues

Simulated Alerts

Users may encounter a problem if they set simulated alerts and then save the state of the Vega Position Sensor. The Vega firmware provides the ability for the user to set simulated alerts that the Vega Position Sensor will then send back to the host computer. This allows the software developer to check that their software handles the alerts correctly. In combined firmware revision 002, a problem may occur if the user sets a simulated alert and then saves the state of the Vega Position Sensor. When the Save operation is performed after a simulated alert is set, the simulated alert persists and is treated as if it was a real alert until the alert is cleared. This does not create a fault in the system.

Marker Wavelength

Customers who use active wired tools with ROM files that do not define the marker wavelength should note that for such tools, the Polaris Vega System assumes that the marker wavelength is 880 nm. The previous version of the system (the Polaris Spectra) assumed that active wired tools were built with 930 nm markers if the wavelength was not specified in the ROM file. The Polaris Vega only supports 880 nm markers. Users need to ensure that they use only NDI 880 nm active markers with Vega. Failure to do so may result in inaccurate data being returned by the system.

Fail Safe Mode

The Polaris Vega Position Sensor's front panel LEDs do not indicate when the Vega Position Sensor is in fail safe mode. If used, the System Control Unit makes no LED indication either. The Polaris Vega System includes fail safe firmware which takes over in the event that the main firmware failed to boot. The fail safe firmware provides minimal functionality that is sufficient to load new firmware into the Polaris Vega, but does not provide tracking functions. This may occur if a firmware update was interrupted (for example by a power interruption to the system) and did not complete successfully. In this case you should repeat the firmware update. This may also occur if the Polaris Vega System did not boot successfully twice in a row (for example, if the power was interrupted during two successive boot attempts). In this case, rebooting the Polaris Vega one more time and allowing it to complete the boot cycle will clear the problem.

When the system is in fail safe mode,

- The network connection method for the device is always set to DHCP+Zeroconfig.
- The ToolBox **Connect to** menu shows the device is in fail safe mode, as shown on page 60.
- On the network, the device host name changes to "PSU" or "SCU" as appropriate.
- When connected to the device, the VER 4 command returns "Polaris Vega Loader Firmware" or "Polaris Vega SCU Loader Firmware.

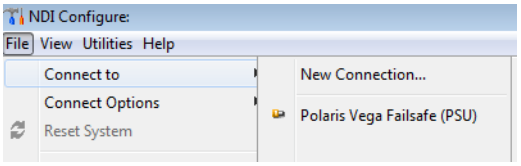


Figure 9-1 Connect to menu in Failsafe mode

BX Commands

If a BX command (such as BX 0001) is used, then the expected transformations are not returned from the Vega Position Sensor unless the `Info.Status.New Alerts` user parameter is read prior to requesting transformations with the BX command. BX 0801 does not experience this problem.

10 Polaris Vega and Lyra Boot and Safe Loader

Note **The Polaris Vega and Lyra boot and safe loader packages are used to update the system's boot loader and safe loader components. Both components are normally invisible to the user.**

The boot loader is a small piece of software that, after power-up, performs a basic initialization of the system and starts the actual booting process of the firmware. As soon as the firmware boot has started, the boot loader will be overwritten.

The purpose of the safe loader is to provide the user with a minimal API to allow for firmware updates. The safe loader becomes active only after the system has gone through two unsuccessful boot attempts. Reasons for unsuccessful boot attempts could be:

- Corrupt firmware
- Premature interruption of boot (for example, the system is unplugged during boot)
- Failed firmware update (the update process was interrupted)

The process for getting the system back into working order depends on the reason why the system ended up in safe loader mode in the first place. In particular, if the system ended in safe loader mode because of premature interruption of the boot process, then a simple reboot will get the system back to proper operation. If the system ended in safe loader mode because of corrupt firmware or a failed firmware update, then a successful firmware update must be performed before the system can boot into proper operational mode again.

When the Position Sensor is in fail safe mode:

- The amber error status LED on the front panel will blink.
- The alert 0x00010000 is raised, indicating the firmware is running in fail safe mode.
- The reply to the ver 4 command will contain the phrase “Loader Firmware” as compared to “Control Firmware”
- In versions 003 and 004 of the Polaris Vega safe loader, if you use mDNS, the system will announce itself as “Polaris Vega Failsafe” instead of “Polaris Vega”. In addition, the system serial number will appear as P9-UNPROGRAMMED instead of the actual serial number of the Position Sensor.
- In version 006 and later, a Polaris Vega Position Sensor announces itself as “Polaris Vega” with the correct serial number of the Position Sensor.
- In version 010 and later, a Polaris Lyra Position Sensor announces itself as "Polaris Lyra" with the correct serial number of the Position Sensor.

11 Abbreviations and Acronyms

Acronym or Abbreviation	Definition
DTLS	Datagram Transport Layer Security
FPGA	Field Programmable Gate Array
GBF	General Binary Format
GPIO	General Purpose Input/Output
IR	Infrared Light
LLDP	Link Layer Discovery Protocol
mDNS	Multicast Domain Name System
NDI	Northern Digital Inc.
PoE	Power Over Ethernet
PSE	Power Sourcing Equipment
PTP	Precision Time Protocol
RTC	Real Time Clock
RTSP	Real Time Streaming Protocol
SCU	System Control Unit
TCP	Transmission Control Protocol
TLS	Transport Layer Security
UDP	User Datagram Protocol